

Mohnish Nanthakumar

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EDUCATION:

University of Wisconsin-Madison **GPA: 3.5**
B.S. Computer Engineering, Computer Sciences

Madison, WI
Expected Graduation: May 2028

WORK EXPERIENCE:

Embedded Software Engineering Intern (Incoming) – Plexus Corp. – C++ *Sep 2026 – Dec 2026*

- Will write **production firmware** for **safety-critical** systems in **modern C++** with RTOS
- Will contribute to a large codebase with **GoogleTest** validation

Embedded Systems Applications Intern – DigiKey – Python, Linux, Bash *May 2026 – Aug 2026*

- **Designed and implemented** robotic sorting system for electronic components on **BeagleBoard** embedded Linux
- Reverse-engineered robotic arm's initialization sequence for programmatic control of industrial 6-DOF arm after diagnosing ROS2 as unviable (220% CPU overhead)
- Implemented coordinate transformation, mapping camera pixel space to robot Cartesian coordinates, enabling millimeter precision when picking up electronic components
- Implemented minimal compute image processing pipeline (threshold/erosion/contour) for detecting components
- **Trained and deployed** TFLite transfer learning model (MobileNet, 1300+ images, 98% test accuracy) on embedded board to classify components from camera captures

TEAM PROJECTS:

Avionics Engineer - Wisconsin Space Program (WiSP) – C++ *Sep. 2024 – Dec 2025*

- **Developed** flight computer and ground station on **ESP32** — real-time telemetry (altitude, acceleration) via LoRa RF with **unit test coverage** for all major functions and documented data flow for maintainability
- **Designed PCB** for Engine Control Unit (**Arduino MEGA**) of bipropellant liquid rocket (IPA/LOX) — reads engine temperature and pressure sensors and transmits to ground station
- Headed the New Member Project to teach electronics through microcontrollers (**Arduino Nano**) and PCB design

Software Engineer – FIRST Robotics Team 3130 (High School Robotics Team) - Java *Sep. 2022 – May 2024*

- **Developed, tested, and tuned** control software for 120lb competition robots within a **3-month deadline**, integrating diverse hardware subsystems via a high-level framework

PERSONAL PROJECTS:

Bare-Metal Alarm Clock – C

- Implemented alarm system on STM32F4 via register-level programming using datasheet and reference manual
- Peripherals: DMA, TIM PWM, ADC (analog dials), UART (MP3 module), GPIO, external interrupts

AWARDS:

PCB Defect Detection - UW Madison Design Lab Makeathon (2nd Place) - Python

- Developed a **Computer Vision** system (Raspberry Pi) to automate detection of defects on PCBs

TECHNICAL SKILLS:

Embedded: C, C++, STM32, ESP32, **BeagleBoard**, **Raspberry Pi**, **Arduino**, FreeRTOS, Bare-Metal, UART, SPI, I2C, DMA, Docker

Software: **Python**, **Linux/Unix**, Bash, Git, GitLab CI, Make, OpenCV, TensorFlow/TFLite, Java

Hardware: **PCB Design**, **Electrical Test Equipment**, MATLAB, Onshape